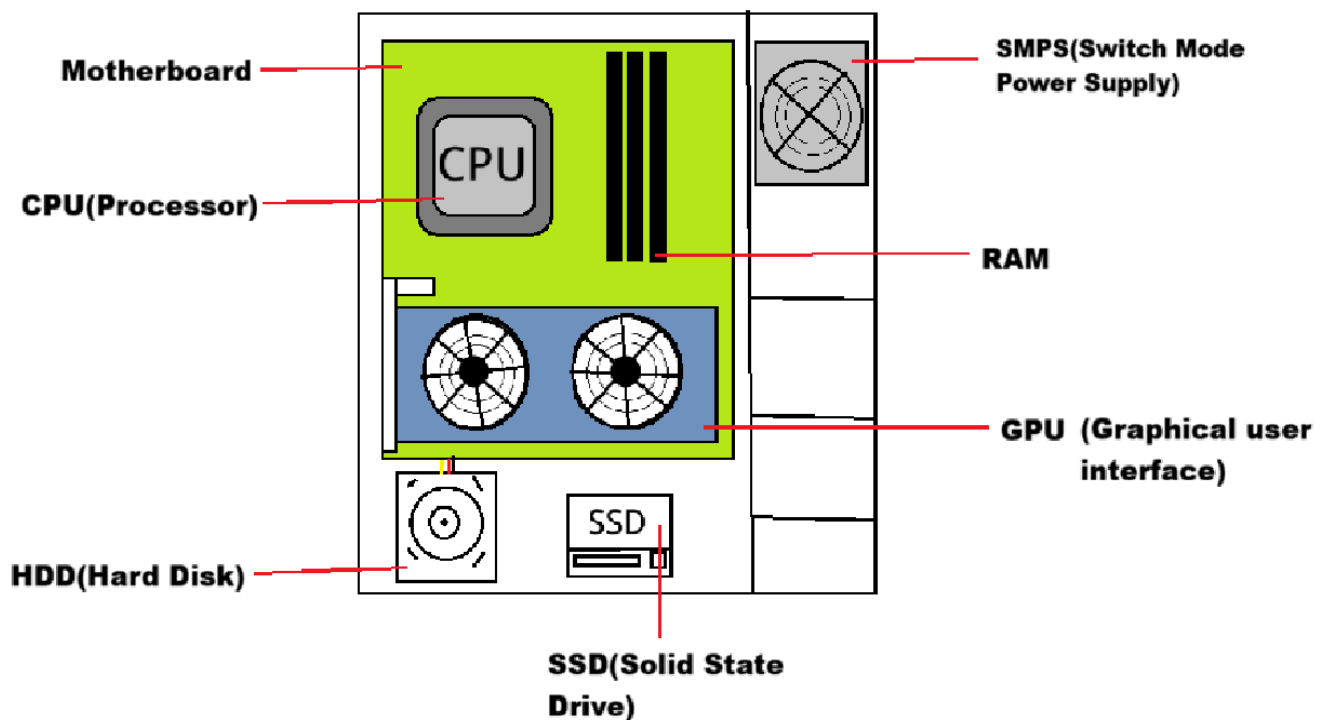


Assignment-1.1

“Hardware and it's Components”

TASK-1

Draw a labelled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



TASK-2

Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature	HDD	SDD
Speed	Slower	Faster
Durability	Less shock-resistant	More shock-resistant
Use Cases	Large storage, backups	Gaming,Laptops

- **HDD(Hard disk)** - HDD is mechanical based storage device, it has a spinning head and tiny arms, data and files are spread across the platter and each tiny arm has to swing to fetch data across the platter. HDD is used to store large static data,and backups.This process takes

time , therefore HDD is slower. HDD has a moving parts so they can get damaged by physical shock.

- **SSD(Solid State Disk)** - SSD is electronic flash memory based storage device , electricity is fast so data is fetched instantly as computer requests the data. SSD has no moving parts , so it doesn't get damaged easily from physical shock or vibrations. It is durable. SSD is widely used in laptops, gaming ,OS and applications.

TASK -3 Write a short explanation (2–3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

- 1. CPU - 'CPU is the brain of the Computer', it does all algorithms and calculation and also process data and give output according to instruction. A stronger CPU can run logic quickly and give output faster, so the frames and character's performance will get update quickly and game will process task smoothly. -If CPU is weak, logic and frames rate will get dropped and your activity will be slower than other players because the computer will not keepup with calculations.**
- 2. RAM - RAM is temporary memory used by the computer while programs are running. Fetching data from SDD/HDD takes milliseconds thus the game runs smoothly. If RAM is not enough, Specially games like PUBG requires massive data to load, If RAM runs slow , maps and dashboard will shutter and walls may visible blank.**
- 3. GPU - The GPU processes and helps creating graphics.It helps create things such as characters, textures, lighting, environments, and visual effects, colour.A powerful GPU can generally allows higher graphics settings and smoother frame rates, depending on the game and the whole computer decktop. If the game becomes slow slideshow or looks blurry because you have to drop the resolution.**

TASK- 4 Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Common Problem	Cause	Solution
Stuttering , Lagging	Game running out of workspace	Upgrade RAM
Long loading screens	Drive reading too slowly	Upgrade SSD
Blurr, half image, low fps	Graphics processor struggling	Upgrade GPU
Game freezing in large number of participants	Processor is being used 100%	Upgrade CPU

- ☐ **Upgrading RAM**- During FIFA , RAM also competes for background processes such as browser, window . RAM ensures FIFA has uninterrupted, dedicated memory space to run smoothly.
- ☐ **Upgrading SSD**- Before match, everything inside FIFA such as assets, stadium model, commentary files, player information, rule book must be load before match starts. Switching from HDD to SDD, loading time of screens reduces much lesser than before.
- ☐ **Upgrading GPU** - Games like FIFA has realistic properties like players exact face, differents angles , dimentions, colors, shadows, perfect lighting. New GPU processes these High Resolution visual very smoothly.
- ☐ **Upgrading CPU**- FIFA has many scoreboards which requires statistics and Graphs to display on the dashboard also collective score from previous level. This process requires accurate and fast calculation while working on other desktop instruction. Upgraded CPU will help to achieve smoother and effective FIFA experience.